Experiment Title

Study of LAN Topology, Network Devices, and Address Configuration.

Problem Statement

1) To study the LAN topology used in College campus.
2) To identify networking device & understand their function.
3) To analyze wired & wireless Communication method used in the network.
4) To measure speed of Network for wired & wireless connections.
5) To obtained MAC, IP Address & Subnet mask.

Draw & identify LAN topology of College Campus & Clearly label all Components such as router, Switches & etc.

Sol^n :- The College Campus network observed this study Consists of :

- Layer-3 core switches in the Server room.

- managed by POE + switches.
- Wireless Access point (APs)
- Ethernet port in classroom & Laboratories.

→ Network Topology

Hybrid Hierarchical Topology which Combines :

1) Tree Topology
2) Extended Star Topology
3) Ring Topology.

→ The Network hierarchical layer

i) Core layer
ii) Distribution layer
iii) Access layer

→ • Core layer :-

Core layer Contains Server Room and Server room have Firewall, Router, 2- L3+ Switches, Patch Rack, Server, inverter, etc.

• Distribution Layer :-

• Distribution layer Contains 4-

- Switches.

  - The 4 Switches are :-

    i) main Build -> R7 (Room no - 213, 2nd Floor)
    ii) main Build -> R1 (Room no - 301, 3rd Floor)
    iii) main Build -> R6 (Room no - 113, 1st Floor).
    iv) main Build -> R2 (Room no - 201, 2nd Floor).

- This 4 - switches Contains ->

  ① 18 Port Gigabit Easy Smart Switch with 16 - Port. $\boxed{\text{TL - SG1218MPE}}$

  ② $\boxed{\text{SG 3428 XMP}}$ -> 24 Port Gigabit & 4-port 10 GE SFP+, L2 + managed switches with 24 Port POE+ .

- Now Let's discuss these 2 above Component :-

① <u>TP - Link TL - SG1218MPE</u>

- Likely used for : CCTV Cammera, wifi access point, etc.

- 18 Port Gigabit "Easy Smart" switch.

- 16 Port Support POE+

② <u>TP - Link Omada SG 3428 XMP</u>

- Higher - end managed Layer 2 + Switch.
- 24 POE + Port, uplinks, 10G SFP
- Usually acts as a main distribution for each Floor.

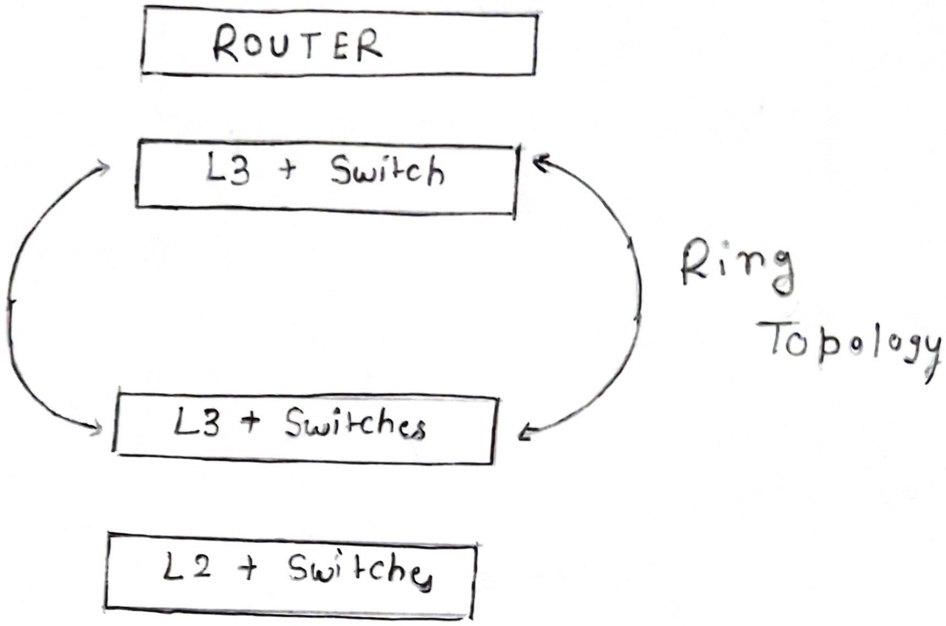
Access Layer

- Access layer Contain 3 - Switches i.e
  1) main Build R1 - [Room no - 301, 3rd Floor].
  2) main Build R4 - [Room no - 05, Ground "].
  3) main Build R5 - [Room no - 18, Ground Floor].

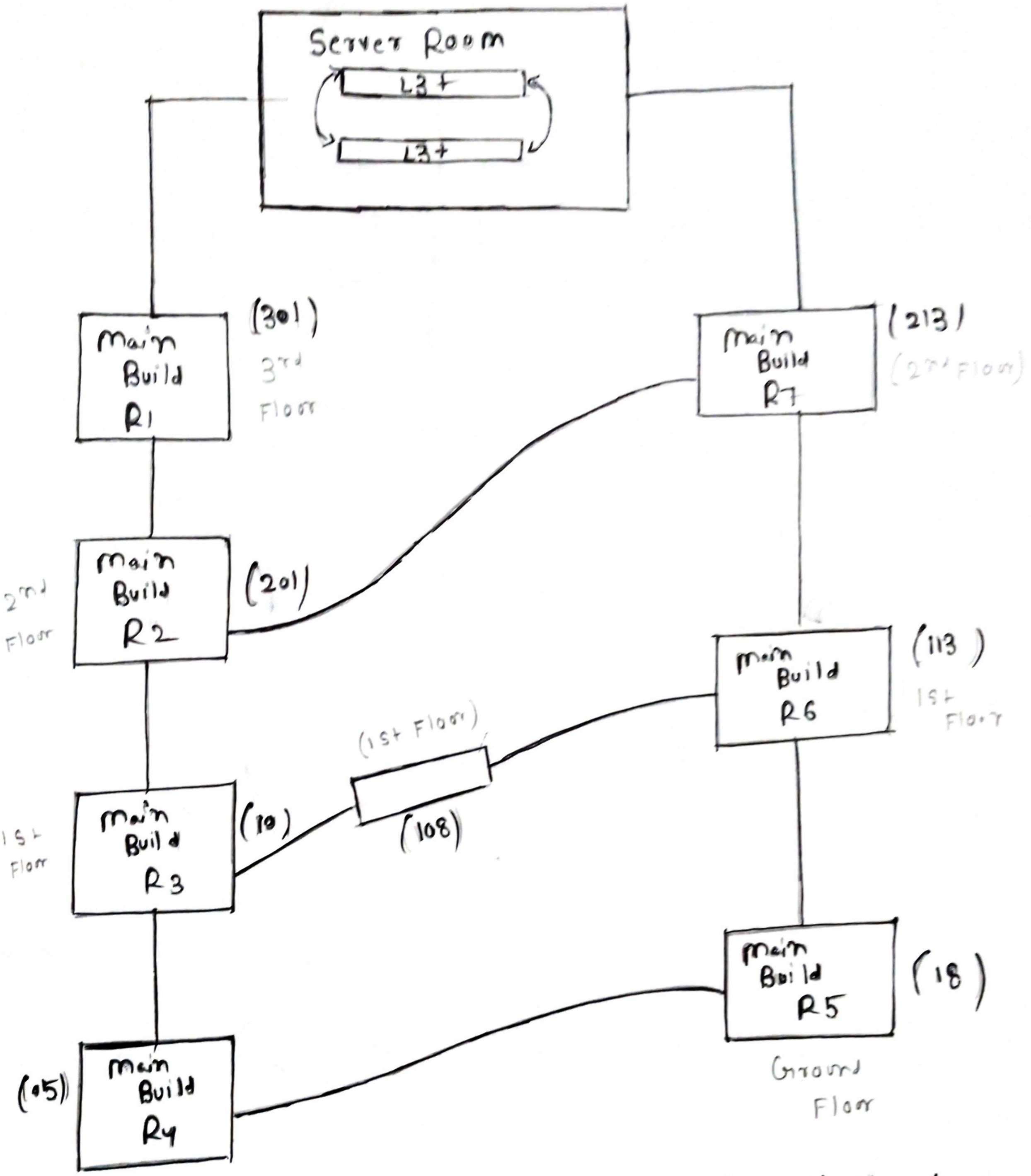
- Every switches act as access layer which helps to provide network Connectivity to all classroom, Lab, Faculty Room, etc.

- Access Layer directly Connect:
  1) Ethernet Port
  2) wireless AP
  3) Lab System
  4) Outdoor wireless AP (ex- 018)

→ Server Room Architecture / Topology

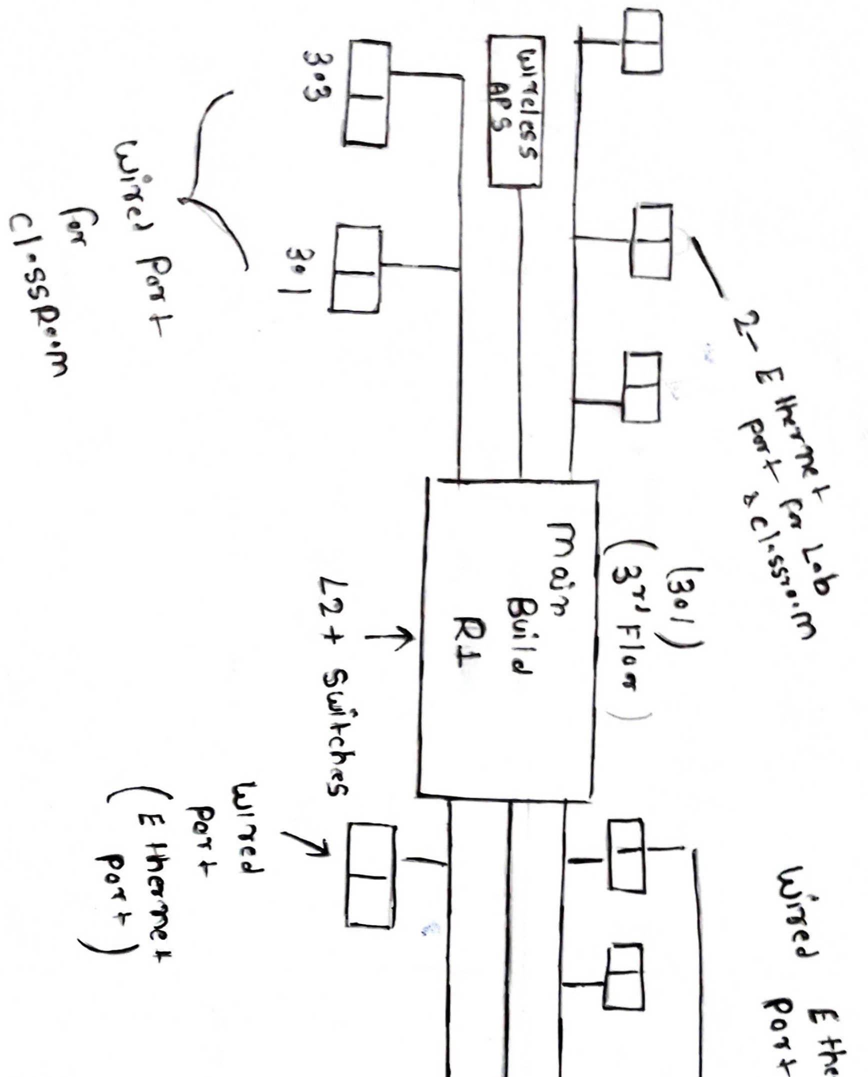


Topology



- → R1, R7, R2, R6 → Distribution layer
- → Server room → Core layer
- → R1, R2, R3, R4, R5, R6, R7 → ACCESS layer

Access Layer Switch (3rd Floor - 301)



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Windows PowerShell
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PS C:\Users\ADMIN> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : hgu_lan

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::992c:6cd2:c438:a76c%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:7724:848f:bc56:1acc:5293:d899
   Temporary IPv6 Address. . . . . . : 2401:4900:7724:848f:d9ca:8701:255f:333d
   Link-local IPv6 Address . . . . . : fe80::ec27:de91:8697:173c%5
   IPv4 Address. . . . . . . . . . . : 10.240.176.129
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::105e:c4ff:fec4:afac%5
                                       10.240.176.182

PS C:\Users\ADMIN> |
```

→ Using "IP config"

WIRELESS CONNECTION

```
PS C:\Users\ADMIN> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::992c:6cd2:c438:a76c%8
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3e84:595f:7e3a:56c9%15
   IPv4 Address. . . . . . . . . . . : 10.85.19.63
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 10.85.16.1

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\ADMIN>
```

WIRED CONNECTION

→ Using IPCONFIG

WIRELESS SPEED

Performance

Run new task

CPU
5% 0.81 GHz

Memory
5.8/7.8 GB (74%)

Disk 0 (C:)
SSD
0%

Ethernet
Ethernet 2
S: 0 R: 0 Kbps

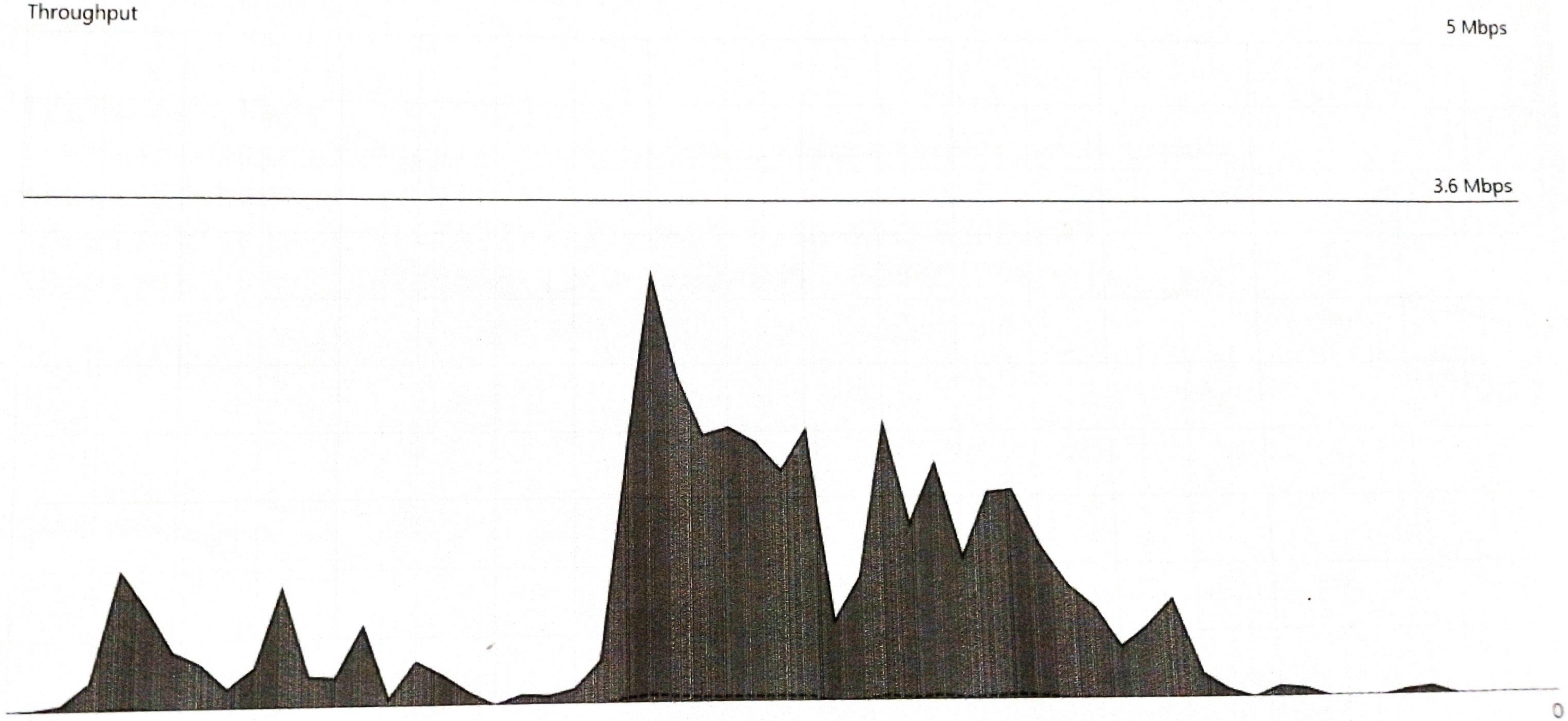
Wi-Fi
Wi-Fi
S: 0 R: 0 Kbps

GPU 0
Intel(R) HD Graphics 6...
1%

Wi-Fi

Intel(R) Dual Band Wireless-AC 8265

Throughput



Send
0 Kbps

Receive
0 Kbps

Adapter name:	Wi-Fi
SSID:	Galaxy A038755
Connection type:	802.11n
IPv4 address:	10.81.18.68
IPv6 address:	2409:408a:1cc4:21b2:d49c:34a5:75bb:69af
Signal strength:	

↳ MAC A

```
Windows PowerShell        ×   +  ⌄

Windows PowerShell
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PS C:\Users\ADMIN> getmac

Physical Address    Transport Name
=================== ==========================================================
B0-0C-D1-2C-3A-DF   Media disconnected
0E-6B-7F-E1-99-D2   \Device\Tcpip_{27635621-3133-4B8B-9255-E7C4BB9E0EA7}
0A-00-27-00-00-08   \Device\Tcpip_{3F28BB86-23C7-4F82-8497-6D44C1776D07}
PS C:\Users\ADMIN>
```

→ MAC ADDRESS

→ using "getmac"

→ Difference b/w wired & wireless speed ?

Solⁿ

Wired Network	Wireless Network
i) Faster Communication	i) Slower Communication
ii) Suuitable for tasks requiring high band-width	ii) Suitable for mobile devices and general usage.
iii) Low latency	iii) Higher latency
iv) Stable Connection.	iv) Affected by distance/ interference.
v) Dedicated physical medium	v) Shared radio medium.

- Conclusion

Wired Connections are faster and more flexiable while wireless Connections offer flexiobility and easy to use.

→ Role of Subnet mask in network

- A subnet mask plays an important role in network communication.

- It divides an IP address in network part and host part.
- Helps in identifying whether a device belongs to the same network or not.
- Enables effeicient routing of data within a network.
- Reduces network congestion by organi-zing devices into subnets.

Example

If the IP Address is 192.168.1.5 and Subnet mask is 255.255.255.0,

- Network portion -> 192.168.1
- Host Portion -> 5

→ Conclusion :

The analysis shows that while wired network provide better speed and stability, wireless networks are more convenient, Additionally subnet masks are crucial for efficient data transmission and network management